Dr. RVR
NRI
INSTITUTE OF TECHNOLOGY DEEMED TO BE
UNIVERSITY

LEARN
INNOVATE
EXCEL
TRANSFORM

DR. RVR NRI INSTITUTE OF TECHNOLOGY

(DEEMED TO BE UNIVERSITY) • AGIRIPALLI, ANDHRA PRADESH, INDIA

NRI
INSTITUTE OF TECHNOLOGY DEEMED TO BE
UNIVERSITY

MINI PROJECT REPORT

PROJECT REPORT TEMPLATE & RESEARCH QUESTIONNAIRE

Project Title : ____________________
Student Name(s) : ____________________
Roll Number(s) : ____________________
Department : ____________________
Project Guide : ____________________
Academic Year : 2026 – 2027

NRI
INSTITUTE OF TECHNOLOGY
DEEMED TO BE UNIVERSITY
AGIRIPALLI, ANDHRA PRADESH

[PROJECT TITLE]

Student Name: [Your Name]

Roll Number: [Your Roll Number]

Department: [Department Name]

College: [College Name]

Project Guide: [Guide Name]

Abstract

[Write a concise summary of the problem, proposed approach, dataset, main results, and conclusion. Replace this instruction with your own content.]

Keywords

[Keyword 1]; [Keyword 2]; [Keyword 3]; [Keyword 4]; [Keyword 5]

1. Problem Statement and Objectives

1.1 Problem Statement

1.2 Objectives

2. Literature Review (Research Gap)

Review relevant research papers, preferably covering the years 2021–2026. Summarize each paper accurately and identify the gap your project addresses.

Year	Model / Method	Author(s) / Paper	Limitations / Research Gap
[Enter details]	[Enter details]	[Enter details]	[Enter details]
[Enter details]	[Enter details]	[Enter details]	[Enter details]
[Enter details]	[Enter details]	[Enter details]	[Enter details]
[Enter details]	[Enter details]	[Enter details]	[Enter details]
[Enter details]	[Enter details]	[Enter details]	[Enter details]
[Enter details]	[Enter details]	[Enter details]	[Enter details]

Research gap identified: [Briefly explain the unresolved issue your project addresses.]

3. Block Diagram and Explanation

[Insert the system architecture / block diagram here.]

Figure 1. [Caption for the block diagram]

Explanation of the major modules and data flow

4. Dataset Used

Dataset name and source

Number of samples / images / records and classes

Input features and data description

Preprocessing and data cleaning

Train / validation / test split (if applicable)

Dataset limitations or constraints

5. Proposed Model / Methodology

Proposed model / algorithm

Step-by-step working procedure

Tools, programming languages, and libraries

Why this approach was selected

6. Results

6.1 Experimental Setup

6.2 Performance Evaluation Metrics

6.3 Results Obtained

6.4 Performance Analysis and Observations

7. Graphs

7.1 Graphical Representation of Results

7.2 Comparison with Existing Methods (if applicable)

7.3 Confusion Matrix / Performance Comparison Graphs (as applicable)

7.4 Discussion and Interpretation of Graph

8. Conclusion and Future Scope

8.1 Conclusion

8.2 Future Scope

9. References

List 10 relevant research papers or credible sources used in your project. Use one consistent citation style and verify every reference.

[1] Author(s), “Paper Title,” Journal / Conference, Year, DOI or URL.

[2] Author(s), “Paper Title,” Journal / Conference, Year, DOI or URL.

[3] Author(s), “Paper Title,” Journal / Conference, Year, DOI or URL.

[4] Author(s), “Paper Title,” Journal / Conference, Year, DOI or URL.

[5] Author(s), “Paper Title,” Journal / Conference, Year, DOI or URL.

[6] Author(s), “Paper Title,” Journal / Conference, Year, DOI or URL.

[7] Author(s), “Paper Title,” Journal / Conference, Year, DOI or URL.

[8] Author(s), “Paper Title,” Journal / Conference, Year, DOI or URL.

[9] Author(s), “Paper Title,” Journal / Conference, Year, DOI or URL.

[10] Author(s), “Paper Title,” Journal / Conference, Year, DOI or URL.

MINI PROJECT QUESTIONNAIRE (Q&A)

SECTION I: INTRODUCTION

1.1 **Research Background**

1. **Student Name(s) and Roll Number(s):**
2. **Project Title:**
3. **Project Domain / Research Area:**
4. **What is the background of your research topic?**
5. **Why is this problem important in the selected application domain?**
6. **What real-world problem does your project address?**

1.2 **Research Gap and Motivation**

7. **What are the limitations of existing systems or research methods?**
8. **What specific research gap did you identify from the literature?**
9. **Why did you select this research problem?**
10. **What motivates you to develop the proposed solution?**

1.3 **Research Objectives**

11. **What is the primary objective of your project?**
12. **List the specific objectives of your research.**
13. **What are the expected outcomes of the project?**

1.4 **Major Contributions**

14. **What are the major contributions of your proposed work?**

15. **What is new or improved compared with existing approaches?**
16. **What benefits are expected from your proposed system?**

SECTION II: RELATED WORK

2.1 **Existing Methods**

2.2 **Critical Analysis of Existing Methods**

19. **Which algorithms or methods have been used in previous research?**

20. **What are the strengths of these existing methods?**

21. **What are their limitations or shortcomings?**

22. **Which existing method will you use as a baseline for comparison, and why?**

2.3 **Research Gap**

23. **What specific gap remains unresolved in the existing literature?**

24. **How does your proposed work address this gap?**

25. **What evidence from the reviewed papers supports your identified research gap?**

SECTION III: PROBLEM FORMULATION

3.1 **Problem Definition**

26. **Clearly define the problem your project aims to solve.**

27. **What are the inputs to your proposed system?**

28. **What outputs or predictions are expected?**

29. **Who are the intended users or beneficiaries of your solution?**

3.2 **Mathematical Formulation**

30. **Have you formulated the research problem mathematically?**
- **Yes**
- **No**
- **In Progress**
- **Not Applicable**

31. **If applicable, provide the mathematical equations, objective function, or loss function used.**

32. **Define the important variables, parameters, and symbols used in your formulation.**

3.3 **Research Constraints and Assumptions**

33. **What assumptions are made in your research?**

34. **What technical, computational, or data-related constraints exist?**

35. **What are the scope and boundaries of your proposed solution?**

SECTION IV: PROPOSED METHODOLOGY

4.1 **System Architecture**

36. **Have you prepared a system architecture or block diagram?**
- **Yes**
- **No**
- **In Progress**

37. **Upload or attach the system architecture diagram, if available.**

38. **Explain the major modules and their functions.**

39. **Describe the flow of data between the modules.**

4.2 **Proposed Algorithm**

40. **Which algorithm(s) or machine learning model(s) are used?**

41. **Is the algorithm newly developed, modified, or an existing algorithm?**
- **Newly developed**

- **Modified existing algorithm**
- **Existing algorithm applied to a new problem**
- **Combination of existing algorithms**
- **Not Applicable**

42. **Explain the step-by-step working procedure of your proposed algorithm.**

43. **What advantages does it offer over the selected baseline method?**

4.3 **Mathematical Model**

44. **What mathematical model supports your proposed methodology?**

45. **What equations, optimization techniques, or model parameters are used?**

46. **How does the mathematical model help solve the identified problem?**

4.4 **Algorithm Workflow**

47. **Describe the complete workflow from input data to final output.**

48. **Have you prepared a flowchart or algorithm pseudo code?**

- **Yes**
- **No**
- **In Progress**
- **Not Applicable**

49. **What programming language, libraries, and development tools are used?**

SECTION V: EXPERIMENTAL SETUP AND EVALUATION

5.1 **Dataset Description**

50. **What dataset is used in your project?**

51. **What is the dataset source?**

- **Publicly available dataset**
- **Institution-provided dataset**
- **Self-collected dataset**
- **Synthetic/simulated dataset**

- **Other**

52. **How many records, images, samples, or observations are included?**

53. **How many features or input variables are used?**

54. **What preprocessing techniques are applied?**

55. **Have you addressed missing values, class imbalance, and data leakage?**

5.2 **Experimental Setup**

56. **What hardware and software environment is used?**

57. **How is the dataset divided into training, validation, and test sets, where applicable?**

58. **What model parameters or hyper parameters are used?**

59. **Have you documented the experimental procedure sufficiently for reproducibility?**

5.3 **Evaluation Metrics**

60. **Which evaluation metrics are used to assess your model?**
- **Accuracy**
- **Precision**
- **Recall**
- **F1-score**
- **Mean Absolute Error (MAE)**
- **Root Mean Square Error (RMSE)**
- **AUC-ROC**
- **Execution time / computational cost**
- **Other appropriate metric**

61. **Why are these metrics suitable for your research problem?**

62. **Have you completed the experiments?**
- **Yes**
- **No**
- **Partially**

SECTION VI: RESULTS AND DISCUSSION

6.1 **Performance Evaluation**

63. **What results have you obtained from your proposed system?**

64. **Provide the actual measured values for the selected evaluation metrics.**
65. **What are the major observations from your experimental results?**

6.2 Comparative Analysis

66. **Have you compared your proposed method with existing algorithms or baseline models?**
- **Yes**
- **No**
- **In Progress**
- **Not Applicable**

67. **List the baseline models used for comparison.**

68. **Provide a comparative results table containing the measured results of your proposed method and baseline models.**

69. **What improvement, if any, does your method demonstrate over the baselines?**

70. **Are the comparisons conducted under consistent experimental conditions?**

6.3 Ablation Study

71. **Does your proposed system contain multiple components whose individual contributions can be evaluated?**
- **Yes**
- **No**
- **Not Applicable**

72. **Have you conducted an ablation study?**
- **Completed**
- **In Progress**
- **Planned**
- **Not Applicable**

73. **If applicable, which components were removed or modified, and what effect did this have on the results?**

6.4 Limitations and Discussion

74. **What are the major limitations of your proposed system?**

75. **What challenges or unexpected results occurred during experimentation?**

76. **What factors may affect the reliability or generalization of your results?**
77. **What additional experiments are required to strengthen your findings?**

SECTION VII: CONCLUSION AND FUTURE WORK

7.1 **Conclusion**

78. **What are the major findings of your project?**

79. **To what extent have you achieved your stated research objectives?**

80. **What is the main contribution of your proposed work?**

81. **What practical applications could benefit from your findings?**

7.2 **Future Research Directions**

82. **What improvements can be made in future versions of your project?**

83. **Can the proposed method be tested on larger or different datasets?**

84. **What additional algorithms, models, or techniques could be investigated?**

85. **What further experiments are required before submitting a research paper?**

FINAL PROJECT STATUS AND GUIDE REVIEW

86. **Current project status:**
- **Problem identification**
- **Literature review**
- **Methodology design**
- **Implementation in progress**
- **Implementation completed**
- **Experimental evaluation completed**
- **Dissertation/report preparation**

87. **Have you prepared the following research artifacts?**
- **Literature review with verified references**
- **Research gap and objectives**

- **Problem formulation**
- **System architecture diagram**
- **Algorithm and mathematical model, if applicable**
- **Dataset and experimental setup**
- **Results and comparative analysis**
- **Ablation study, if applicable**
- **Limitations and future work**
- **Complete dissertation/report**

88. **Do you intend to develop your project into a research paper?**

- **Yes**
- **No**
- **Need guidance from project guide**

89. **What additional support do you require from your project guide?**

STUDENT DECLARATION AND GUIDE REVIEW

Student Declaration: I confirm that the information provided is accurate and reflects the actual progress of my project. All results and references will be reported truthfully.

Student Signature: ______________________________________

Project Guide Name: ____________________________________

Guide Remarks: ____

Guide Signature: ____ Date: _______________

Dr. RVR
NRI
INSTITUTE OF TECHNOLOGY DEEMED TO BE
UNIVERSITY

ENGINEERING IDEAS
FOR A BETTER TOMORROW

EDUCATION | RESEARCH | INNOVATION

Dr. RVR
NRI
INSTITUTE OF TECHNOLOGY DEEMED TO BE
UNIVERSITY

Dr. RVR NRI Institute of Technology
(Deemed to be University)
Agiripalli, Andhra Pradesh, India